

State and Proof Green's Theorem

Statement:- If 'R' is a closed region of the xy-plane bounded by a simple closed curve 'C'. If 'M' and 'N' are continuous functions of 'x' and 'y'. Then we have to show that

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Proof:- Consider $\vec{P}(x,y) = M(x,y)\hat{i} + N(x,y)\hat{j}$
Also $\vec{r}(x,y) = x\hat{i} + y\hat{j}$ (In xy-plane)

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

Now we have $\vec{P}(x,y) \cdot d\vec{r} = M(x,y)dx$

$$\int_C \vec{P}(x,y) d\vec{r} = \int_C M(x,y) dx$$

$$\therefore \int_C \vec{P}(x,y) d\vec{r} = \int_{c_1} M(x,y) dx + \int_{c_2} M(x,y) dx$$

$$\int_C \vec{P}(x,y) d\vec{r} = \int_a^b M(x, y_1(x)) dx + \int_b^a M(x, y_2(x)) dx$$

$$\int_C \vec{P}(x,y) d\vec{s} = \int_a^b M(x, y_1(x)) dx - \int_a^b M(x, y_2(x)) dx$$

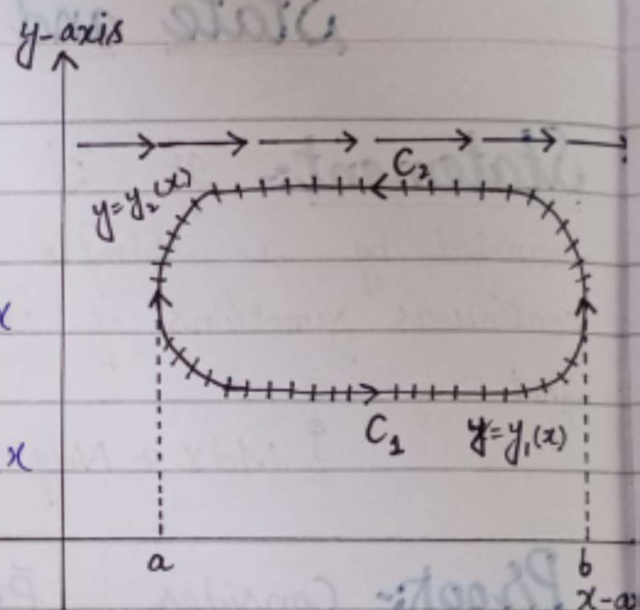
$$\int_C \vec{P}(x,y) d\vec{s} = \int_a^b [M(x, y_1(x)) - M(x, y_2(x))] dx$$

$$\int_C \vec{P}(x,y) d\vec{s} = - \int_a^b [M(x, y_2(x)) - M(x, y_1(x))] dx$$

$$\oint_C \vec{P}(x,y) d\vec{s} = - \int_a^b \left. \frac{\partial M}{\partial y} \right|_{y_1(x)}^{y_2(x)} dx$$

$$\oint_C \vec{P}(x,y) d\vec{s} = - \int_a^b \int_{y_1(x)}^{y_2(x)} \frac{\partial M(x,y)}{\partial y} dy dx$$

$$\oint_C \vec{P}(x,y) d\vec{s} = - \iint_R \frac{\partial M}{\partial y} dx dy \longrightarrow (1)$$



Also; $\vec{P}(x,y) d\vec{s} = \langle N(x,y) dy$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_C N(x,y) dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_{C_1} N(x,y) dy + \int_{C_2} N(x,y) dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_b^a N(x_2(y), y) dy + \int_a^b N(x_1(y), y) dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = - \int_a^b N(x_2(y), y) dy + \int_a^b N(x_1(y), y) dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_a^b [N(x_1(y), y) - N(x_2(y), y)] dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_a^b \left. \frac{\partial N(x,y)}{\partial x} \right|_{x_2(y)}^{x_1(y)} dx dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \int_a^b \int_{x_2(y)}^{x_1(y)} \frac{\partial N}{\partial x} dx dy$$

$$\oint_C \vec{P}(x,y) d\vec{s} = \iint_R \frac{\partial N}{\partial x} dx dy \longrightarrow (2)$$

Adding eq(1) and eq(2)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

This completes the proof.

State and Proof Stoke's Theorem

Statement: If 'S' is an open, two sided, continuous and non intersecting curve 'C' and ' $\vec{A}$ ' is a continuous derivative then $\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$ where $\hat{n}$ is traversed in the positive direction.

Proof:- If 'S' is an open, two sided, continuous and non-intersecting curve 'C'. then $z = f(x, y)$, $x = g(y, z)$, $y = h(x, z)$ is a single valued, continuous and non-intersecting curve 'C'.

$$\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k} \longrightarrow (1)$$

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k} \longrightarrow (2)$$

Consider

$$\nabla \times (A_1 \hat{i}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} = \frac{\partial A_1}{\partial z} \hat{j} - \frac{\partial A_1}{\partial y} \hat{k}$$

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS = \left[\frac{\partial A_1}{\partial z} \hat{j} - \frac{\partial A_1}{\partial y} \hat{k} \right] \cdot \hat{n} dS$$

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS = \left[\frac{\partial A_1}{\partial z} \cdot \hat{n} \cdot \hat{j} - \frac{\partial A_1}{\partial y} \cdot \hat{n} \cdot \hat{k} \right] dS \quad (3)$$

If $z = f(x, y)$ is considered to be the equation of 'S'. Since the position vector

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

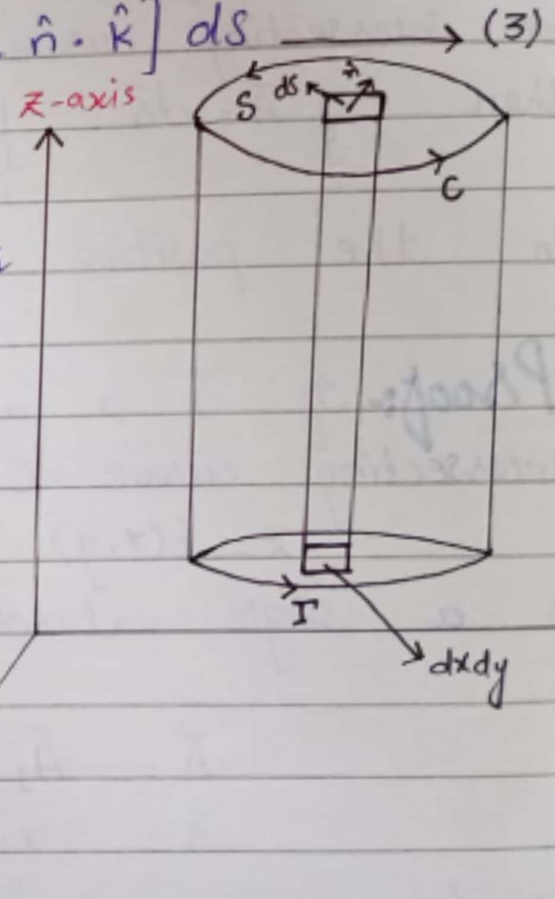
$$\vec{h} = x\hat{i} + y\hat{j} + f(x,y)\hat{k}$$

$$\frac{\partial \vec{v}}{\partial y} = 0 + \hat{j} + \frac{\partial z}{\partial y} \hat{k}$$

$$\frac{\partial \vec{r}}{\partial y} = \hat{j} + \frac{\partial z}{\partial y} \hat{k} \longrightarrow (4)$$

Since $\frac{\partial \vec{r}}{\partial y}$ is vector tangent to S .

So, it is perpendicular to $\hat{n}$.



$$\hat{n} \cdot \frac{\partial \hat{r}}{\partial y} = 0$$

$$\hat{n} \cdot \left(\hat{j} + \frac{\partial z}{\partial y} \hat{k} \right) = 0$$

$$\hat{n} \cdot \hat{j} + \frac{\partial z}{\partial y} \hat{n} \cdot \hat{k} = 0$$

$$\hat{n} \cdot \hat{j} = - \frac{\partial z}{\partial y} \hat{n} \cdot \hat{k} \longrightarrow (5)$$

Using this in eq (3)

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} ds = \left[\frac{\partial A_1}{\partial z} \left(- \frac{\partial z}{\partial y} \hat{n} \cdot \hat{k} \right) - \frac{\partial A_1}{\partial y} \hat{n} \cdot \hat{k} \right] ds$$

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} ds = \left[- \frac{\partial A_1}{\partial z} \cdot \frac{\partial z}{\partial y} - \frac{\partial A_1}{\partial y} \right] \hat{n} \cdot \hat{k} ds$$

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} ds = - \left[\frac{\partial A_1}{\partial y} + \frac{\partial A_1}{\partial z} \cdot \frac{\partial z}{\partial y} \right] \hat{n} \cdot \hat{k} ds \longrightarrow (6)$$

On S , $A_1(x, y, z) = A_1(x, y, f(x, y)) = F(x, y)$, So,

$$\frac{\partial F}{\partial y} = \frac{\partial A_1}{\partial x} \cdot \frac{\partial x}{\partial y} + \frac{\partial A_2}{\partial y} \cdot \frac{\partial y}{\partial y} + \frac{\partial A_3}{\partial z} \cdot \frac{\partial z}{\partial y}$$

$$\frac{\partial F}{\partial y} = \frac{\partial A_1}{\partial y} + \frac{\partial A_3}{\partial z} \cdot \frac{\partial z}{\partial y} \longrightarrow (7)$$

Using this in eq (6) we have

$$[\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS = - \frac{\partial F}{\partial y} dx dy \quad \therefore \hat{n} \cdot \hat{k} dS = dx dy$$

$$\iint_S [\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS = \iint_R - \frac{\partial F}{\partial y} dx dy \longrightarrow (8)$$

Where R is the projection of S on the xy -plane. By the Green's theorem for the plane, we have.

$$\iint_R - \frac{\partial F}{\partial y} dx dy = \oint_{\Gamma} F dx \longrightarrow (9)$$

Where Γ is the boundary of R . Since at each point (x, y) Γ the value of 'F' is same as the value of 'A₁'.

$$\oint_C F dx = \oint_C A_1 d\vec{x} \longrightarrow (10)$$

Putting this value in eq(9)

$$\iint_R - \frac{\partial F}{\partial y} dx dy = \oint_C A_1 d\vec{x} \longrightarrow (11)$$

Putting this value in eq(8)

$$\iint_S [\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS = \oint_C A_1 \cdot d\vec{x} \longrightarrow (12)$$

Similarly, the projections on the other planes, we have

$$\iint_S [\nabla \times (A_2 \hat{j})] \cdot \hat{n} dS = \oint_C A_2 \cdot d\vec{y} \longrightarrow (13)$$

$$\iint_S [\nabla \times (A_3 \hat{k})] \cdot \hat{n} dS = \oint_C A_3 \cdot d\vec{z} \longrightarrow (14)$$

Adding eq(12), eq(13) and eq(14)

$$\iint_S [\nabla \times (A_1 \hat{i})] \cdot \hat{n} dS + \iint_S [\nabla \times (A_2 \hat{j})] \cdot \hat{n} dS + \iint_S [\nabla \times (A_3 \hat{k})] \cdot \hat{n} dS =$$

$$\oint_C A_1 dx + \oint_C A_2 dy + \oint_C A_3 dz$$

$$\iint_S [\nabla \times (A_1 \hat{i}) + \nabla \times (A_2 \hat{j}) + \nabla \times (A_3 \hat{k})] \cdot \hat{n} dS = \oint_C A_1 dx + \oint_C A_2 dy + \oint_C A_3 dz$$

$$\iint_S [\nabla \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})] \cdot \hat{n} dS = \oint_C (A_1 + A_2 + A_3) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\iint_S (\nabla \times \vec{A}) \cdot \hat{n} dS = \oint_C \vec{A} \cdot d\vec{r}$$

$$\Rightarrow \oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \hat{n} dS$$

This completes the proof.

Gauss's Divergence Theorem

Statement:- If V is the volume bounded by a closed surface S and $\vec{A}$ is a vector function of position with continuous derivatives, then

$$\iiint_V \nabla \cdot \vec{A} dV = \iint_S \vec{A} \cdot \hat{n} dS$$

where $\hat{n}$ is the positive (outward drawn) normal to S .

Proof:- Let S be a closed surface which is such that any line parallel to the coordinate axes cuts S in at most two points. Assume the equations of the lower and upper portions S_1 and S_2 to be $z = f_1(x, y)$ and $z = f_2(x, y)$ respectively. Let the projection of the surface on the xy -plane be denoted by R , if

$$\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k} \longrightarrow (1)$$

Then
$$\iiint_V \frac{\partial A_3}{\partial z} dV = \iiint_V \frac{\partial A_3}{\partial z} dz dy dx = \iint_R \left[\int_{f_1(x,y)}^{f_2(x,y)} \frac{\partial A_3}{\partial z} dz \right] dy dx = \iint_R [A_3]_{z=f_1}^{z=f_2} dy dx$$

$$\Rightarrow \iiint_V \frac{\partial A_3}{\partial z} dV = \iint_R [A_3(x, y, f_2) - A_3(x, y, f_1)] dy dx \longrightarrow (2)$$

Since the unit normal $\hat{n}_2$ to S_2 makes an acute angle γ_2 (makes an obtuse angle)^{*} with $\hat{k}$ for the upper portion S_2 . So, $dydx = \cos \gamma_2 dS_2 = \hat{k} \cdot \hat{n}_2 dS_2$. Using this for upper portion, we have

$$\iint_R A_3(x, y, f_2) dy dx = \iint_{S_2} A_3 \hat{k} \cdot \hat{n}_2 dS_2 \longrightarrow (3)$$

Similarly; Since the unit normal $\hat{n}_1$ to S_1 makes an obtuse angle γ_1 with $\hat{k}$ for the upper portion S_1 . So, $dydx = -\cos \gamma_1 dS_1 = -\hat{k} \cdot \hat{n}_1 dS_1$. Using this for lower portion, we have,

$$\iint_R A_3(x, y, f_1) dy dx = -\iint_{S_1} A_3 \hat{k} \cdot \hat{n}_1 dS_1 \longrightarrow (4)$$

Subtracting eq(4) from eq(3)

$$\iint_R A_3(x, y, f_2) dy dx - \iint_R A_3(x, y, f_1) dy dx = \iint_{S_2} A_3 \hat{k} \cdot \hat{n}_2 dS_2 + \iint_{S_1} A_3 \hat{k} \cdot \hat{n}_1 dS_1$$

$$\iint_R [A_3(x, y, f_2) - A_3(x, y, f_1)] dy dx = \iint_S A_3 \hat{k} \cdot \hat{n} dS \quad \because \hat{n} = \hat{n}_2 + \hat{n}_1$$

Putting this in eq(2)

$$\iiint_V \frac{\partial A_3}{\partial z} dV = \iint_S A_3 \hat{k} \cdot \hat{n} dS \longrightarrow (5)$$

Similarly;

$$\iiint_V \frac{\partial A_2}{\partial y} dV = \iint_S A_2 \hat{j} \cdot \hat{n} dS \longrightarrow (6)$$

And;

$$\iiint_V \frac{\partial A_1}{\partial x} dV = \iint_S A_1 \hat{i} \cdot \hat{n} dS \longrightarrow (7)$$

Adding eq (7), (6) and (5) we have.

$$\iiint_V \frac{\partial A_1}{\partial x} dV + \iiint_V \frac{\partial A_2}{\partial y} dV + \iiint_V \frac{\partial A_3}{\partial z} dV = \iint_S A_1 \hat{i} \cdot \hat{n} dS + \iint_S A_2 \hat{j} \cdot \hat{n} dS + \iint_S A_3 \hat{k} \cdot \hat{n} dS$$

$$\iiint_V \frac{\partial A_1}{\partial x} dV + \frac{\partial A_2}{\partial y} dV + \frac{\partial A_3}{\partial z} dV = \iint_S [A_1 \hat{i} \cdot \hat{n} dS] + [A_2 \hat{j} \cdot \hat{n} dS] + [A_3 \hat{k} \cdot \hat{n} dS]$$

$$\iiint_V \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) dV = \iint_S (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \cdot \hat{n} dS$$

$$\oint_V \nabla \cdot \vec{A} dV = \oint_S \vec{A} \cdot \hat{n} dS$$

This completes the proof.

Statement in words:-

The surface integral of the normal component of a vector $\vec{A}$ taken over a closed surface is equal to the integral of the divergence of $\vec{A}$ taken over the volume enclosed by the surface.

